

CS 4700/6700 Introduction to Database Management Systems Fall, 2026

Description: Study of logical and physical aspects of database management systems. Entity-Relationship (ER) and relational models for databases are presented. Relational algebra operations, SQL queries, physical database design methods, and operating principles of database systems are discussed.

Prerequisite: CS 3100 Data Structures and Algorithms, or equivalent.

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Class: MWF, 11:15 am–12:10 pm, 146 Russ Center

Office hour: MWF, 12:30–1:30 pm. For short questions, please use emails.

Text Book: R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, 7th edition, Addison-Wesley, 2016.

Topics:

- DBMS concepts and architecture (Chap 1, 2)
- Entity-Relationship (ER) model (Chap 3)
- Relational data model (Chap 5)
- ER to Relational mapping (Sec. 9.1)
- Relational algebra (Chap 8)
- SQL – a relational database language (Chap 6, 7)
- File organizations and hashing (Chap 16)

Grading: A:[85,100], B:[75,85), C:[65,75), D:[55,65), F:[0,55)

1. There is no homework, but solutions of some exercise questions will be provided.
2. 1st Midterm exam: 15% (10/7, Wednesday, 11:15 am – 12:10 pm)
3. 2nd Midterm exam: 15% (11/18, Wednesday, 11:15 am – 12:10 pm)
4. Final exam: 40% (12/11, Friday, 10:15 am – 12:15 pm)
5. Project: 30%
 - Either (i) a DB design and SQL programming, or (ii) programming of Extendible hashing. Select one and email me your choice by 10/23.
 - Grading of project report (due on 12/4):
 - Database design and SQL programming: {description of problem 4%, ER design 6%, relational DB schema 5%, SQL queries and results 8%, discussion 7%}.
 - Programming of Extendible hashing: {design 5%, documentation 5%, correctness 15%, discussion 5%}.